Classification of Urinary Crystal Images with Deep Learning Models

Project description

The crystals found in urine can be used to establish diagnoses for the detection of numerous diseases. The analysis of these crystals can be carried out using spectrum analyzers, but also through image recognition. These tests are used to detect diseases. By illuminating the crystals with polarized light (under a microscope), images are obtained that highlight the structure of the crystals, making it possible to distinguish them (by a human expert). The use of AI and deep learning is therefore a logical approach in this field to assist the human expert and improve the reliability of their decisions.

This project involves laboratories (Chemistry, Computer Science), the University Hospital Center (CHU) of Orléans, as well as a company that is also interested in studying an automatic method. For the PRI, it will be necessary to become familiar with the data, explore methods for automatic annotation of images and other preprocessing techniques, then implement an initial architecture using deep learning, and finally identify the difficulties and areas for improvement.

Motivation

Urine crystals can be diagnostic markers for numerous diseases. Traditional analysis relies on spectrometers and human experts using polarized light microscopy. However, manual diagnosis is time-consuming and subject to variability. Artificial Intelligence and Deep Learning offer promising methods to standardize results and improve diagnostic reliability.

Objectives

- Study the state of the art (image recognition of urine crystals).
- Collect and preprocess data (urine crystal images).
- Explore annotation/segmentation tools to build a labeled dataset.
- Propose and implement a deep learning model for classification.
- Evaluate and improve the model with medical experts.
- Deliver the trained Deep Learning model in Google Colab

Technologies and Tools

- Programming: Python
- Libraries: TensorFlow/PyTorch, OpenCV, Scikit-learn
- Annotation tools: LabelImg, Label Studio (to be considered)
- Environment: Google Colab, GitLab

Tasks and deliverables

Phase 1 – Literature review and data preparation – duration 1 month

- Review research papers (Google Scholar, PubMed).
- Understand existing datasets.
- Preprocess data: normalize metadata, build structured CSV.
- Study annotation tools and select one.

Deliverables: Preprocessed dataset + CSV

Phase 2 – Dataset creation and annotation – duration 2 months

- Apply annotation tool on sample images.
- Test segmentation/annotation quality.
- Create labeled dataset validated by doctors.

Deliverables: Annotated dataset, annotation workflow note

Phase 3 – Model training – duration 1 month

- Select deep learning architecture (CNN, ResNet – to be considered).
- Train on labeled dataset.
- Identify difficulties and improve iteratively.

Deliverables: Model + training scripts, report with relevant metrics

Phase 4 – Evaluation and demonstration – duration 2 weeks

- Evaluate model with doctors.
- Improve based on feedback.

Deliverables: Final trained model, demo, final report + presentation.